NAME _____ CLASS _____

BWAILA CLUSTER MOCK EXAMINATIONS

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/I

Wednesday, 25th May, 2022 **Time Allowed: 2 hours**

PAPER I

(100 marks)

Instructions

1. **This paper contains 13 printed pages. Please check.**
2. Fill in your **name** at the top of each page.
3. Answer **all** the 20 questions in this paper
4. Use of electronic calculators is allowed.
5. The maximum number of marks for each answer is indicated against each question.
6. In the table provided on this page, **tick** against the number of the question you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

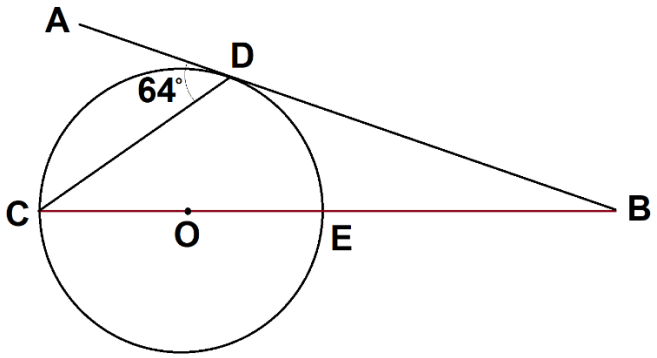
Turn over

1. Factorise completely $(x-y)^2 - y^2$ **(3 marks)**

2. Given that $g(x) = 3x^2$ and $g(a+1) = 3$. Calculate the value of x **(5 marks)**

3. Simplify $201_{10} + 244_5$ giving your answer in base 5 **(5 marks)**

4. Figure 1 shows a circle CDE with centre O. AB is a tangent to a circle at D. BC a diameter



If angle **ADC** $= 64°$, Calculate angle **ABC** **(6 marks)**

5. Express as a single fraction $\frac{1}{x-1} + \frac{2}{xy-y}$ **(3 marks)**

6. Given that $(x-1)$ and $(x-2)$ are both factors of $x^3 + ax^2 + bx - 6$. Find the values of a and b. **(7 marks)**

7. Figure 2 shows unshaded region P bounded by four inequalities

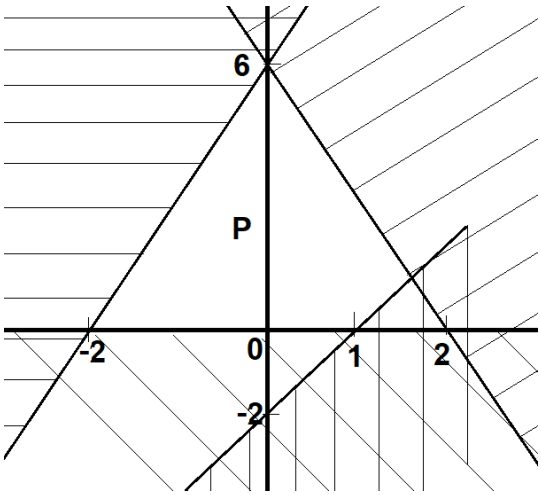


Figure 2

Write down the four inequalities that define region **P**. **(7 marks)**

8. Find the gradient of a straight line whose equation is $\frac{y+2x}{4} = \frac{x}{3}$ **(4 marks)**

9. Given that $A = \sqrt{y(x-b)}$, make b the subject of the formula **(4 marks)**

10. The following is a frequency distribution table showing grades of a test marked out of 30.

Grades obtained	1-5	6-10	11-15	16-20	21-25
No. of students	3	5	6	7	4

Calculate the mean grade scored **(5 marks)**

11. A quantity b varies jointly as r and t when $b = 108 \ when \ r = 3$ and $t = 6$, Find the value of b when $r = \frac{3}{2}$ and $t = 4$ **(6 marks)**

12. In Figure 3, DB is perpendicular to the line AC at point B. AE=15cm, BC=25cm angle EAB=30^0 and angle BCD=45^0

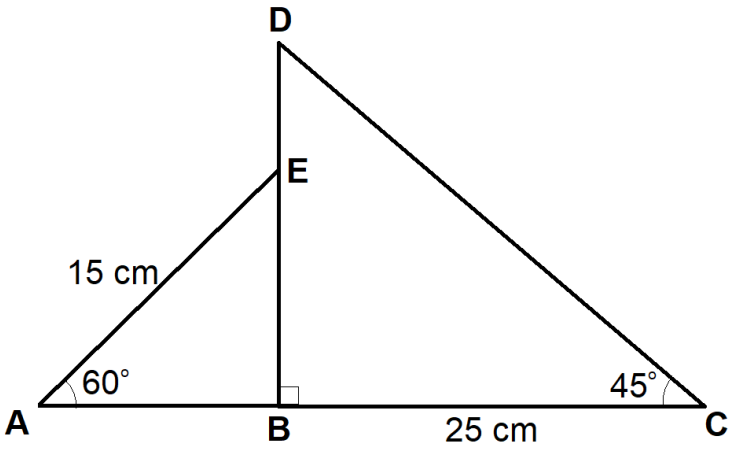


Figure 3

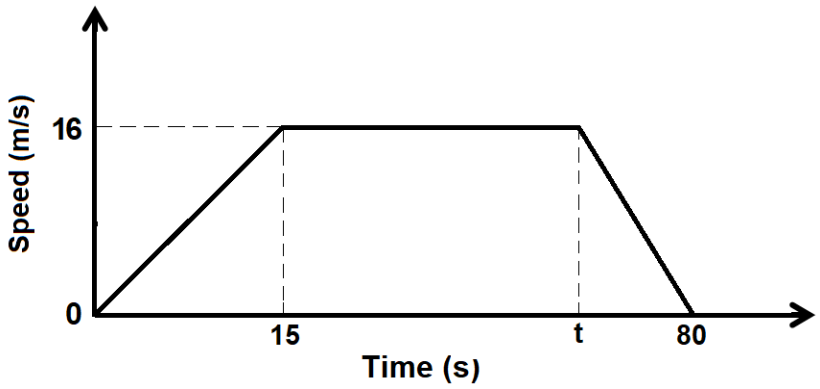
Calculate the length of **DE** leaving your in simplified surd form

(6 marks)

13. The probability that person A will be alive in 20years is $\frac{1}{3}$ and the probability that person B will be alive in 20 years is $\frac{7}{10}$. Use a tree diagram to find the probability that both will be alive in 20 years. **(5 marks)**

14. Matrix $M = \begin{pmatrix} 0 & -10 \\ 2 & 6 \end{pmatrix}$ and $N = \begin{pmatrix} 2 & 5 \\ -4 & 0 \end{pmatrix}$, find $\frac{1}{2}M - N^2$ **(5 marks)**

15. Figure 4 shows a speed-time graph of a train journey



If the total distance travelled is 80 seconds is 920m. find the value of t_2 **(5 marks)**

16. Solve for x in the equation $log_2(x+3) = 2 - log_2 x$ **(5 marks)**

17. A trader sells rice in bags. 30 bags of rice costing K5000 per bag are mixed with 40 bags of another rice costing K5875 per bag. If the mixture is sold at K6600 per bag, calculate the percentage change. **(6 marks)**

18. The 9th term of an Arithmetic Progression $a, a+4, a+8, \ldots is\ 37$. Find the value of a **(4 marks)**

19. Simplify $4^x = 8^{\frac{4}{3}}$ **(4 marks)**

20. **Figure 5** shows circle BCDE with Centre O. Angle BED = 70^0, angle BDC = 37^0 and angle OBE = 25^0

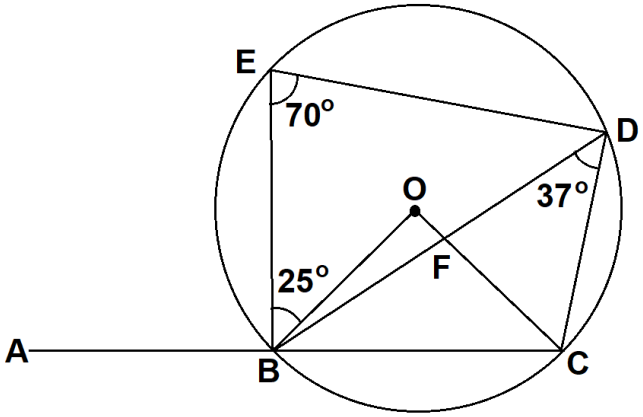


Figure 5

Calculate angle OCD **(5 marks)**

END OF QUESTION PAPER